

VIGYANOTSAV 2.0

2024-25 Brochure

Prepared by: Pehchaan Ek Safar (NGO led by IIT Ropar students)

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WHO ARE WE?

With a team of 150+ active enthusiastic volunteers, Pehchaan-Ek Safar is a registered NGO in Rupnagar, Punjab, founded and run by the students of IIT Ropar. Working with the belief that education is the most important weapon in one's life, the NGO has come a long way. Started with the sole purpose of educating the children living in the slums near and around the IIT Ropar campus, the organization stands strong with several major projects and campaigns meant for the betterment of students' lives.

IIT Ropar provides excellent educational and research facilities. Right outside the campus reside children who lack adequate facilities to get a basic education. A group of students decided to support these kids so that they could grow together and form a sustainable environment both inside and around the campus.

OUR OBJECTIVES

- To create a platform that fosters scientific curiosity and innovation, connecting new ideas with existing knowledge.
- To explore and encourage scientific and technological talent and creative thinking among children.
- To develop an understanding of the role of science and technology in meeting the needs of society.
- To motivate youth that science and technology are instruments for achieving Atmanirbharta in socio-economic development.
- To encourage them as architects of the nation and visualize the future of the nation.

ABOUT VIGYANOTSAV

Pehchaan Ek Safar, an NGO led by IIT Ropar students, is hosting Vigyanotsav, a multifaceted science event aimed at inspiring young minds across Punjab. Vigyanotsav brings together brilliant minds to foster discussions on various scientific topics that have been gaining momentum in recent years.

The main highlight of this year's Vigyanotsav will be a scientific exhibition, inviting students from grades 6 through 12. Students can showcase their talents in various fields of science, including Information and Communication Technology, Life Sciences, Sustainable Energy, and Physical and Mathematical Sciences—though the sky's the limit when it comes to innovation.

In addition to the exhibition, there will be scientific stalls for visitors, featuring games, exhibits on major research topics pursued by experts at IIT Ropar, and much more. It will be a fun-filled experience for all the students who will get to meet professionals from diverse fields and enhance their knowledge of developments in various areas.

Vigyanotsav aims not only to celebrate scientific ingenuity but also to cultivate a sense of discovery among participants. It encourages students to explore their interests, collaborate with peers, and articulate their ideas effectively. In doing so, it nurtures essential skills like critical thinking, problem-solving, and innovation.

EXHIBITION THEMES & SUB-THEMES

Theme 1: Information and Communication Technology (ICT)

Explore the use of technology to process and communicate information. Sub-themes include:

- **Robotics:** Designing simple robots to perform specific tasks.
- **Software Applications:** Developing apps or computer programs to solve real-world problems.
- **Internet of Things (IoT):** Creating smart devices communicating using different protocols.
- **Artificial Intelligence:** Creating projects demonstrating how machines learn (e.g. AI models, chatbots).
- **Communication Systems:** Transmitting information (e.g. radio transmitters, internet concepts).

Theme 2: Life Sciences

Study of living organisms and environmental interactions:

- **Health and Nutrition:** Investigating diets or devices monitoring vital health parameters (heart rate, BP).
- **Environmental Conservation:** Biodiversity protection, plastic reduction, preserving species.
- **Biotechnology:** Eco-friendly plastics, biofuels, or using microorganisms to clean oil spills.
- **Agriculture & Food Technology:** Smart irrigation, soil health monitoring, farming efficiency.

Theme 3: Sustainable Energy & Waste to Wealth/Energy

Renewable energy sources and reducing fossil fuel dependency:

- **Solar Power:** Demonstrating solar panels, solar lamps, fans, and panel efficiency.
- **Wind Energy:** Building simple wind turbines to generate electricity.
- **Biofuels:** Fuel from organic materials like algae or kitchen waste.
- **Hydropower & Energy Conservation:** Mini-hydropower generators, energy-efficient homes.

Theme 4: Physical and Mathematical Science

Laws of physics and mathematical applications:

- **Optics & Light:** Periscopes, reflection/refraction, simple microscopes.
- **Electricity & Magnetism:** Circuits, electromagnets, electric generators.
- **Mathematical Models & Mechanics:** Geometry in architecture, statistics, force, motion, pulley systems.

IMPORTANT INSTRUCTIONS & RULES

Project Requirements & Divisions

- Projects registered must be Working or Static models.
- **Division 1:** Class 6th to 8th
- **Division 2:** Class 9th to 12th
- **Team Composition:** 2 to 3 student members per team. Schools can register multiple teams.

Registration & Visitor Fees

- Participant Fee: Rs. 100 per student | Rs. 100 per supervising teacher.
- Visiting Student Fee: Rs. 50 per student | Rs. 100 per teacher.

Judging Criteria

Projects will be evaluated by expert judges based on: Creative Imagination, Scientific Approach, Originality in Working Model, Technical Skills, and Presentation.

Event Protocol Stages

1. **Stage 1:** Register Team and choose theme.
2. **Stage 2:** Submit Project Idea based on selected theme.
3. **Stage 3:** Demo Video Submission (< 3 mins demonstrating model working).

EVENT SCHEDULE & CONTACT DETAILS

Venue: IIT Ropar Campus, Rupnagar, Punjab

Tentative Date: 1st December 2024

Timings: 9:30 AM to 5:00 PM

School Participation Award

Special award for the school registering the maximum number of participating teams!

Contact Information

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